CSP-203: Software Tools & Technologies Lab

Chapter – 6

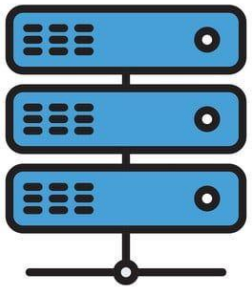
Web Programming : Introduction to Basics (Lecture-1)

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Contents

- ❑ Introduction to Web Development
- ❑ Importance of Web Development
- ❑ Different Types of Development
- ❑ Front-end and Back-end Languages
- ❑ Database
- ❑ Tools
- ❑ Hosting & Deployment
- ❑ Securities
- ❑ APIs
- ❑ Current Trends

Introduction:



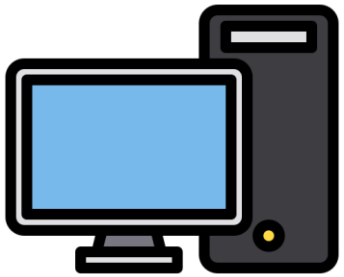
Server





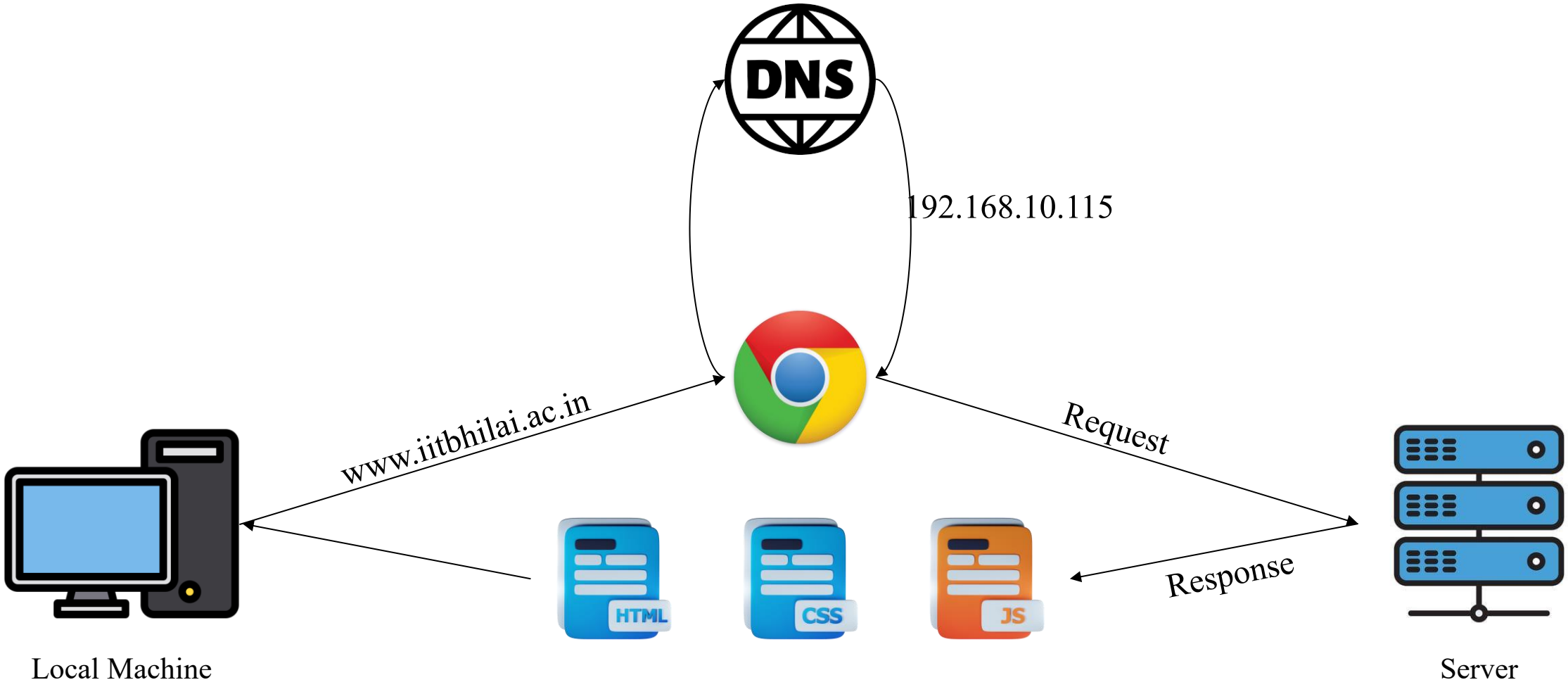






Local Machine

Introduction:



What is Web Development?

- Web development refers to the building, creating, and maintaining websites.
- Involves:
  - Front-End Development: User interface
  - Back-End Development: Server, logic, and database
  - Full-Stack Development: Combination of both

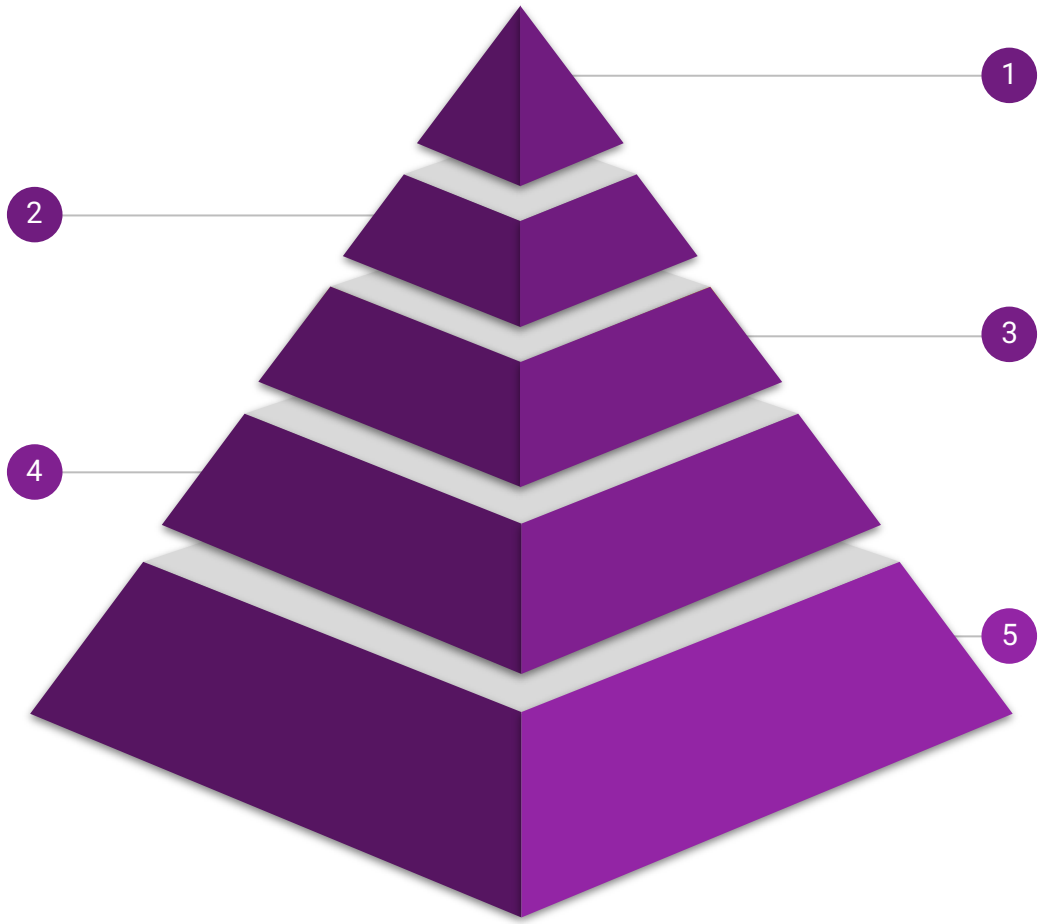
Importance of Web Development:

Accessibility

1. 24*7 Availability
2. Sales Opportunities
3. No Geographical Constraints

Sales & Revenue Boost

1. Attract More Engagement
2. Detailed Information Listing
3. 24*7 Opened Business Store



Business Identity

1. Global Reach
2. Brand’s credibility
3. Visibility

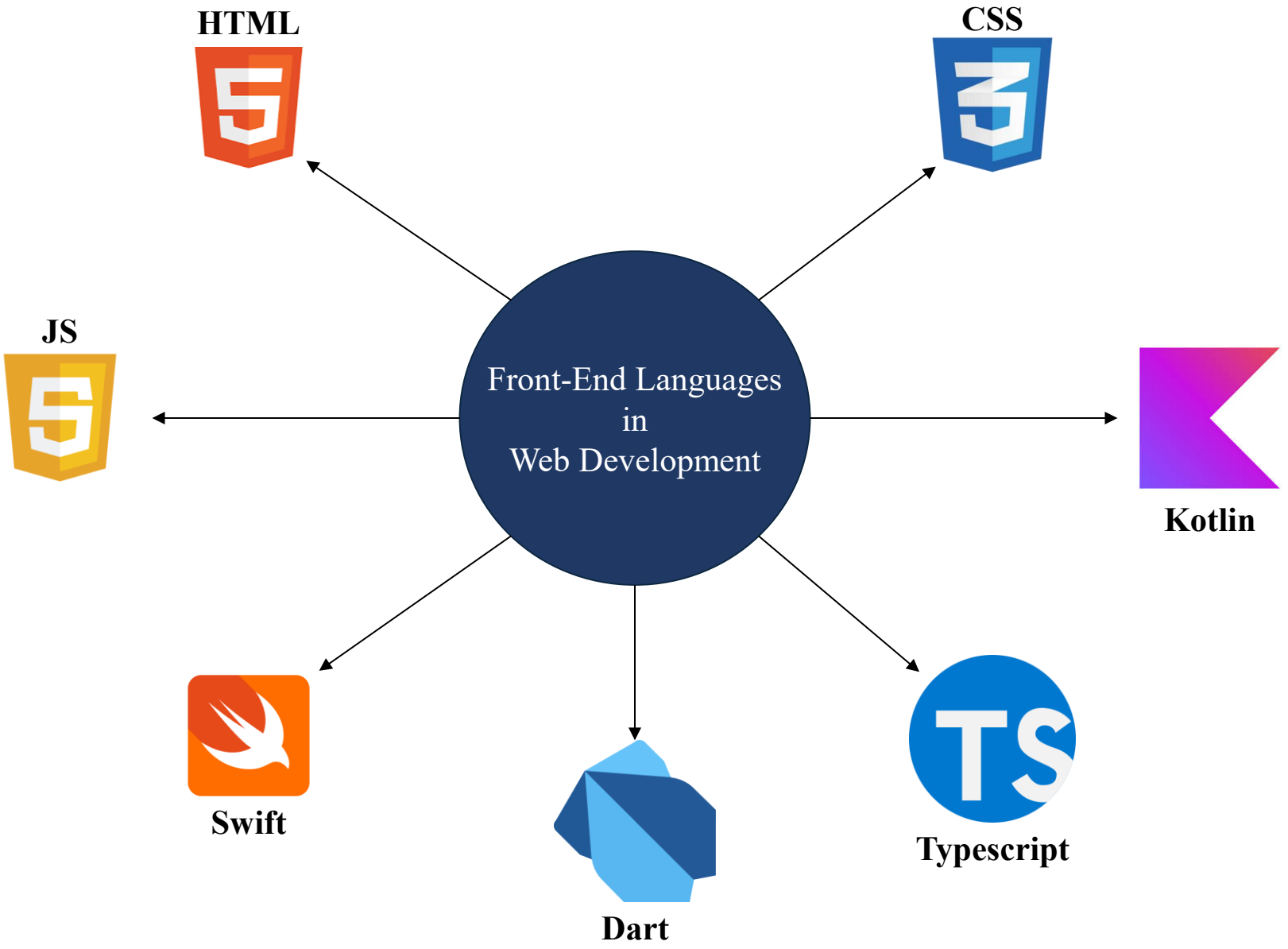
Cost Effectiveness

1. Digital Marketing
2. Search Engine Optimization
3. Low Fixed Cost

Better Customer Experience

1. Easy to Navigate
2. User-friendly
3. No Geographical Constraints

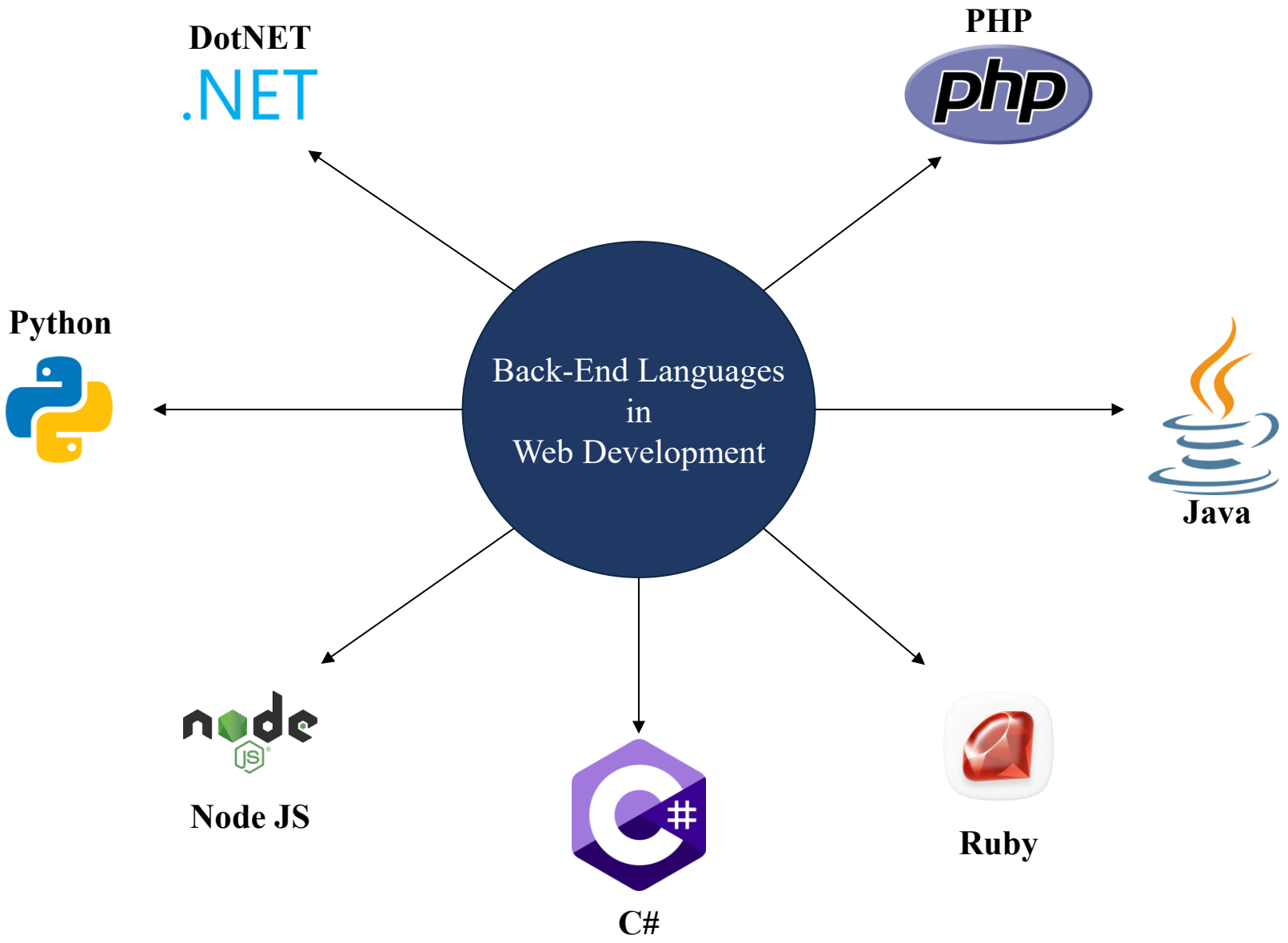
Different Front-End Languages:



Front-End Development:

- Focused on **what users see**.
- Core Technologies:
  - **HTML** (HyperText Markup Language): Page structure
  - **CSS** (Cascading Style Sheets): Styling and layout
  - **JavaScript**: Adds interactivity and logic
- Common Frameworks:
  - **React.js** (by Facebook/Meta)
  - **Angular** (by Google)
  - **Vue.js**
- Importance:
  - Responsive design
  - Great user experience

Different Back-End Languages:



Back-End Development:

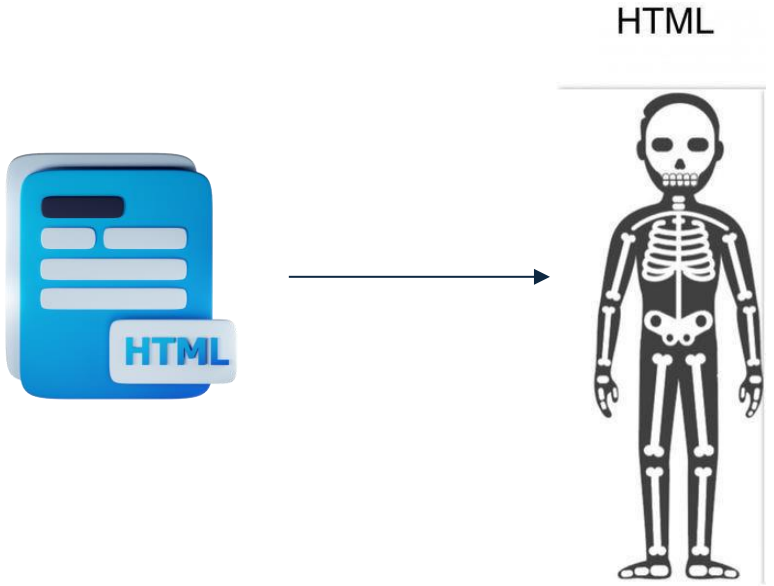
- Controls **server-side logic and databases**.
- Handles:
  - User authentication
  - Database management
  - Application logic
- Languages commonly used:
  - **PHP, Python, Java, C#, Node.js, Ruby**
- Frameworks:
  - **Laravel (PHP), Django (Python), Spring Boot (Java), ASP.NET (C#)**

Full Stack Development:

- Full stack Developers work on both **client and server sides**.
- Understand both front-end and back-end.
- Benefits:
  - Versatile development
  - Independent project handling
- Popular Stacks:
  - **MERN**: MongoDB (non-RDBMS NoSQL database), Express.js (backend framework), React (frontend framework that builds user interfaces), Node.js (open-source backend runtime environment)
  - **MEAN**: MongoDB, Express.js, Angular, Node.js
  - **LAMP**: Linux, Apache, MySQL, PHP

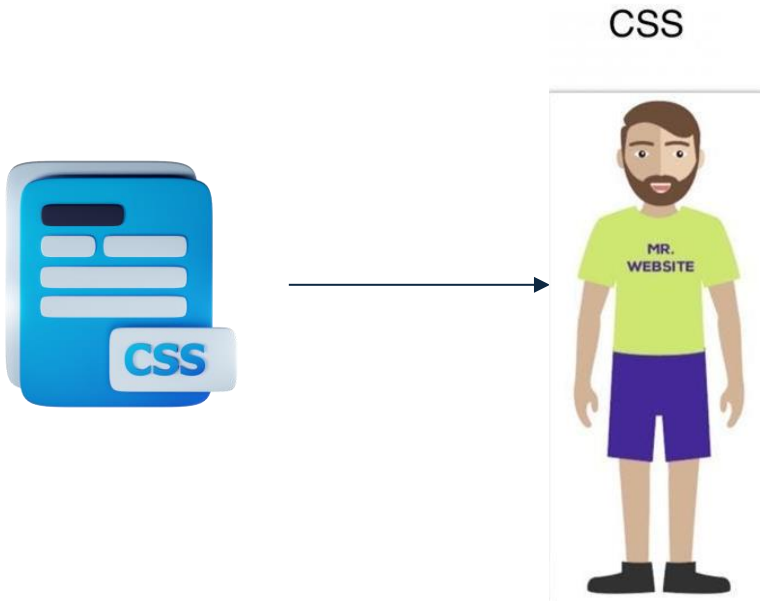
HTML:

- Hyper text markup language
- Standard markup language
- Describes the structure of webpage
- Consists of series of elements
- Elements tell the browser how to display the content.



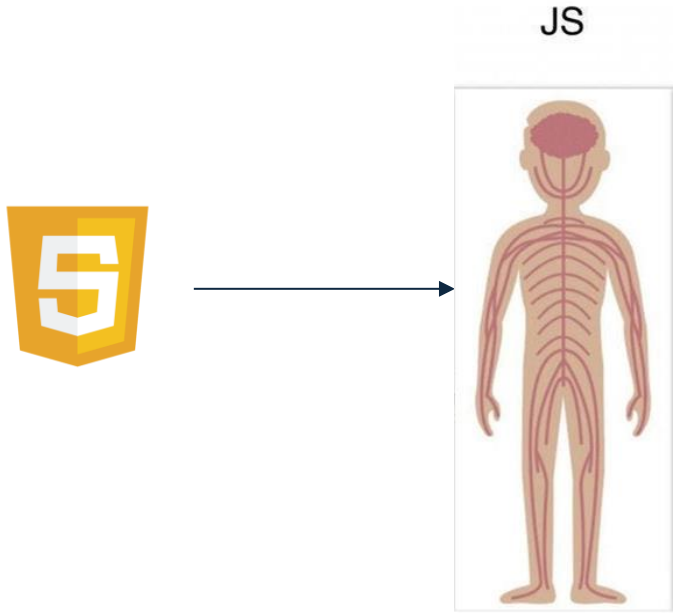
CSS:

- Cascading style sheet.
- Describes how HTML elements are to be displayed on the screen, paper or in other media.
- Saves a lot of work.
- Can control the layout of multiple web pages all at once.



JS:

- Javascript
- A programming language that is one of the core technologies of the World Wide Web, alongside HTML and CSS.
- Add interactivity and logic.



PHP (Hypertext Preprocessor):

- Use Case: Server-side scripting; powering content management systems like WordPress, Joomla, Drupal.
- Importance:
  - One of the oldest and most widely used languages for web back-ends.
  - Powers over 75% of websites (especially via WordPress).
  - Easy to deploy on almost all hosting platforms.
  - Good integration with MySQL and Apache.

Database in Web Development:

- **Store and retrieve** user and app data.
- Types:
  - **Relational Databases (SQL)**: MySQL, PostgreSQL, SQL Server
  - **NoSQL Databases**: MongoDB, Firebase

Web Development Tools:

- **Code Editors**: Visual Studio Code, Sublime Text
- **Version Control**: Git (track changes), GitHub (collaboration)
- **Dev Tools**: Chrome DevTools for testing
- **Deployment**: Docker (containerization), Jenkins (CI/CD)

Hosting & Deployment:

- Hosting = storing your website on a server accessible via the internet.
- Platforms:
  - Shared Hosting: Hostinger, Bluehost
  - Cloud: AWS, Azure, Google Cloud
  - Static Site Hosting: Netlify, Vercel
- Key Concepts:
  - Domain Name
  - DNS Configuration
  - FTP/SSH Access

Securities in Web Development:

- Use HTTPS with SSL certificates
- Validate input on both front-end and back-end
- Prevent common attacks:
  - SQL Injection
  - Cross-Site Scripting (XSS)
  - Cross-Site Request Forgery (CSRF)
- Implement Authentication (login) and Authorization (access control)

APIs in Web Development:

- APIs (Application Programming Interfaces) allow communication between apps.
- Used for:
  - Data exchange
  - Integrating third-party services (e.g., payment gateways)
- Types: (two distinct approaches to designing APIs for data exchange between client and server applications)
  - RESTful APIs
  - GraphQL
- Common Data Format: JSON



HTML

Hypertext Markup Language

Contents

- Introduction
- HTML Tags
- Unordered and Ordered List
- Tables in HTML
- Web Links in HTML
- Images in HTML
- Data Input in HTML
- IFRAME Tags
- Font Tags in HTML
- Using Audio and Video Tags

Introduction

- HTML stands for Hypertext Markup Language.
- It is the standard language for creating web pages.
- It works together with:
  - CSS for styling
  - JavaScript for interactivity
- HTML uses tags (<tag>) to define elements.



Example

- First Example: The basic commands of HTML will be explained by an example.
  - **<!doctype html>** To inform the text editor that the document is an HTML.
  - **<html>** Here the starting of the HTML code, it ends when the </html> is typed.
  - **<head>** The header of the document has the title and the required definitions, such as CSS classes.
  - **<title> Hello </title>** The title shown in the HTML window.
  - **</head>** Closing of the header.
  - **<body>** In body, the text, buttons, and most of tags exist; in this example the body is empty.
  - **</body>** End of body.
  - **</html>** End of HTML code.

Tags

- **<h1>** to **<h6>** – Define headings of different sizes
- **
** – Inserts a single line break
- **<p>** – Defines a paragraph of text
- **<a>** – Anchor tag for hyperlinks
- **<img>** – Embeds an image
- **
** – Line break
- ** ** – Block-level container (used for layout/design)
- **** – Inline container (used for styling text)
- **<ul>**, **<ol>**, **<li>** – Lists (unordered, ordered, and items)
- **<input>** – Form input fields (text, password, etc.)
- **<form>** – Container for form elements
- **<button>** – Clickable button

Tags ends when we close them, otherwise , it will continue. tags used to be nested inside each other. **Note : Tags in the coming examples will be written inside the body.**

Example

```
<!doctype html>
<html>
<head>
  <title> Hello </title>
</head>
<body>
  <p> IIT BHILAI </p> <br>
  <h1> IIT BHILAI </h1> <br>
  <h2> IIT BHILAI </h2> <br>
  <h3> IIT BHILAI </h3> <br>
  <h4> IIT BHILAI </h4> <br>
  <h5> IIT BHILAI </h5> <br>
  <h6> IIT BHILAI </h6> <br>
  <h7> IIT BHILAI </h7> <br>
</body>
</html>
```

What is the output of the program?

Example

```
<!doctype html>
<html>
<head>
  <title> Hello </title>
</head>
<body>
  <p> IIT BHILAI </p> <br>
  <h1> IIT BHILAI </h1> <br>
  <h2> IIT BHILAI </h2> <br>
  <h3> IIT BHILAI </h3> <br>
  <h4> IIT BHILAI </h4> <br>
  <h5> IIT BHILAI </h5> <br>
  <h6> IIT BHILAI </h6> <br>
  <h7> IIT BHILAI </h7> <br>
</body>
</html>
```

What is the output of the program?

IIT BHILAI

IIT BHILAI

IIT BHILAI

IIT BHILAI

IIT BHILAI

IIT BHILAI

Unordered and Ordered Lists

- Used to **arrange multiple items** in a structured way.
- **Ordered list (<ol>)**:
  - Items are automatically numbered (1, 2, 3, ...).
- **Unordered list (<ul>)**:
  - Items are displayed with **bullets** (●) by default.
- Each list item is placed inside a ** tag** (List Item).

Type Attribute in Lists

Ordered List

type value	List Style
1	1, 2, 3, ... (default)
A	A,B,C, ...
a	A,b,c, ...
I	I, II, III, ...
i	i, ii, iii, ..

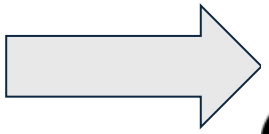
Unordered List

Type value	List Style
disc	● (default)
circle	○
square	■
none	No bullets

Example

```
<h1>Our Departments (Unordered List)</h1>
<ul>
  <li>Computer Science and Enigneering</li>
  <li>Data Science and Artifical Intelligence</li>
  <li>Electronics</li>
</ul>

<h1>Our Departments (Ordered List)</h1>
<ol>
  <li>Computer Science and Enigneering</li>
  <li>Data Science and Artifical Intelligence</li>
  <li>Electronics</li>
</ol>
```



Our Departments (Unordered List)

- Computer Science and Enigneering
- Data Science and Artifical Intelligence
- Electronics

Our Departments (Ordered List)

1. Computer Science and Enigneering
2. Data Science and Artifical Intelligence
3. Electronics

```
<ol type="A">
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ol>
```

Type can be 1,A,a, I/i

Creating Tables in HTML

- Use <table> to define the table.
- Use <tr> to define a table row.
- Use <th> to define header cells (bold by default).
- Use <td> to define normal data cells.
- Tables are created **row-by-row**.
- Use the border attribute to make the table visible.
- Tag structure must be properly closed.

HTML Table Attributes & Tags

S.No	Attribute	Description	Example
1	`border`	Adds border to the table and cells	`<table border="1">`
2	`caption`	Adds a **title** to the table	`<caption>Student Info</caption>`
3	`colspan`	Merges **multiple columns** into one cell	`<td colspan="2">Merged Cell</td>`
4	`rowspan`	Merges **multiple rows** into one cell	`<td rowspan="2">Merged Row</td>`
5	`align` & `valign`	Aligns text **horizontally** and **vertically**	`<td align="center" valign="middle">Text</td>`
6	`width` & `height`	Sets **width or height** of table or cell	`<table width="70%">` / `<td width="100">`
7	`cellpadding`	Adds **space inside each cell**, between content and border	`<table cellpadding="10">`
8	`cellspacing`	Adds **space between cells** in a table	`<table cellspacing="5">`

Example

```
<h2>Student Information Table</h2>

<table border="1">
  <tr>
    <th>Name</th> <!-- Table headers -->
    <th>Age</th>
    <th>Department</th>
  </tr>
  <tr>
    <td>Soham</td> <!-- Table data row 1 -->
    <td>20</td>
    <td>CSE</td>
  </tr>
  <tr>
    <td>Radha</td> <!-- Table data row 2 -->
    <td>21</td>
    <td>EE</td>
  </tr>
</table>
```



Student Information Table

Name	Age	Department
Soham	20	CSE
Radha	21	EE

Example

```
<h2>Student Performance Report</h2>

<table border="1" width="80%" cellpadding="10" cellspacing="5">
  <caption><b>Student Details and Grades</b></caption>

  <tr>
    <th rowspan="2">Name</th>
    <th rowspan="2">Age</th>
    <th colspan="2">Academic Info</th>
    <th rowspan="2">Remarks</th>
  </tr>

  <tr>
    <th>Stage</th>
    <th>Grade</th>
  </tr>

  <tr>
    <td align="center" valign="middle">Soham</td>
    <td align="center">20</td>
    <td align="center">Second</td>
    <td align="center">A</td>
    <td align="center">Passed</td>
  </tr>

  <tr>
    <td align="center" valign="middle">Radha</td>
    <td align="center">21</td>
    <td align="center">Third</td>
    <td align="center">B</td>
    <td align="center">Promoted</td>
  </tr>

  <tr>
    <td colspan="5" align="center"><b>End of Report</b></td>
  </tr>
</table>
```

Student Performance Report

Student Details and Grades

Name	Age	Academic Info		Remarks
		Stage	Grade	
Soham	20	Second	A	Passed
Radha	21	Third	B	Promoted
End of Report				

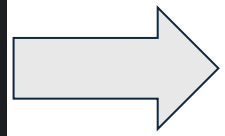
Web Links (<a> Tag)

- The <a> tag is used to **create hyperlinks** in HTML.
- The href attribute specifies the **URL** of the page to link to.
- The **text between the opening and closing <a> tag** is what appears as the clickable link on the webpage.
- Clicking the link **navigates to another page or site**.
- You can link to:
  - External websites (e.g., https://www.example.com)
  - Internal pages (e.g., xyz.html)
  - Email addresses (mailto:someone@example.com)

Example

```
<h2>Visit College Website</h2>

<a href="http://www.iitbhilai.ac.in">
  Click to visit IIT BHILAI Website
</a>
```



Visit College Website

Click to visit IIT BHILAI Website

Adding Images in HTML

- The `<img>` tag is used to **display images** on a webpage.
- It is a **self-closing tag** – no closing `</img>` required.
- **Syntax:**

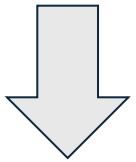
```
<img src="image.jpg" width="300" height="200">
```

- **`src` (source):** Specifies the **file name or path** of the image.
- If no path is provided, the image should be in the **same folder as the HTML file**.
- **`width` and `height`** can be used to adjust the image size in **pixels**.
- You can also use optional attributes like:
  - `alt` – Alternate text shown if image fails to load
  - `title` – Tooltip text when mouse hovers on image

Example

```
<h2>College Logo</h2>

<img src="https://upload.wikimedia.org/wikipedia/en/2/27/IIT_Bhilai_Logo.svg" width="300" height="200" alt="IIT Bhilai Logo" title="IIT Bhilai">
```



College Logo



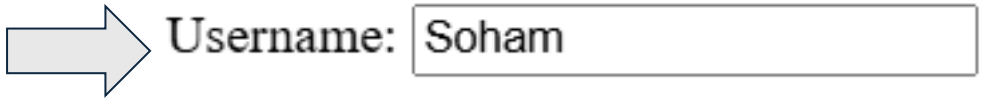
Data Input in HTML

- Use the <form> tag to create a **form container**.
- Inside <form>, use <input> for collecting data.
- Common input types include:
  - text, password, submit, radio, checkbox, number, date, time
- Always pair inputs with **labels** for better accessibility.
- Use name attributes to identify each input field.

1 Text Input Field

➢ Allows user to enter plain text (e.g., name, username).

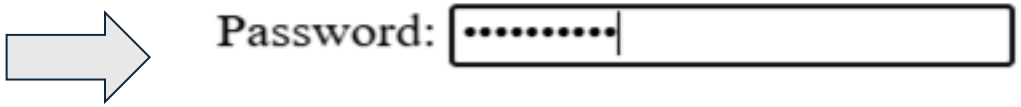
```
<form>
<label for="username">Username:</label>
<input type="text" id="username" name="username">
</form>
```



2 Password Input

- Masks characters with dots or asterisks.

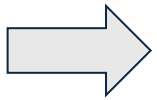
```
<form>
<label for="pwd">Password:</label>
<input type="password" id="pwd" name="pwd">
</form>
```



3 Submit Button

- Submits the form data to the server (action URL).

```
<form>
<input type="submit" value="Submit">
</form>
```

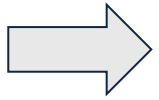


Submit

4 Radio Buttons

- Select one option from multiple choices.

```
<form>
<p>Choose Gender:</p>
<input type="radio" name="gender" value="male"> Male<br>
<input type="radio" name="gender" value="female"> Female<br>
<input type="radio" name="gender" value="Others"> Others<br>
</form>
```



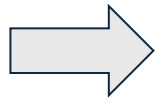
Choose Gender:

- (•) Male
- ( ) Female
- ( ) Others

5 Checkboxes

➢ Select multiple options from a list.

```
<form>
<p>Select your hobbies:</p>
<input type="checkbox" name="hobby" value="reading"> Reading<br>
<input type="checkbox" name="hobby" value="traveling"> Traveling<br>
<input type="checkbox" name="hobby" value="sports"> Sports<br>
</form>
```



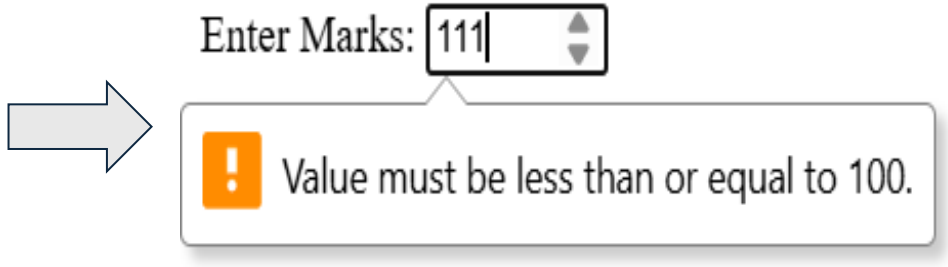
Select your hobbies:

- [x] Reading
- [x] Traveling
- [ ] Sports

6 Number Input

- Only allows numeric input within an optional range.

```
<form>
<label>Enter Marks:</label>
<input type="number" id="marks" name="marks" min="0" max="100">
</form>
```

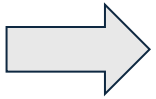


7 Date & Time Input

- Lets users pick a date and time.

```
<form>
<label>Date of Birth:</label>
<input type="date" id="dob" name="dob"><br><br>

<label>Choose Time:</label>
<input type="time" id="appt" name="appt">
</form>
```



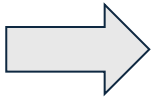
Date of Birth: 03-04-1999

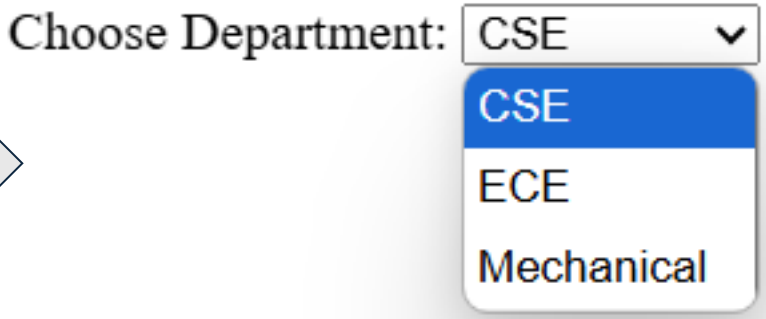
Choose Time: 19:30

8 Dropdown List (Select Box)

- Allows users to select one value from a list.

```
<form>
<label>Choose Department:</label>
  <select id="dept" name="department">
    <option value="cse">CSE</option>
    <option value="ece">ECE</option>
    <option value="mech">Mechanical</option>
  </select>
</form>
```

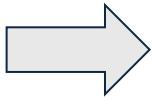




9 Multiline Text Box (Textarea)

- For larger text input (like feedback, address, etc.).

```
<form>
<label>Your Address:</label><br>
<textarea id="address" name="address" rows="5" cols="40"></textarea>
</form>
```



Your Address:

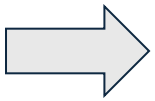
Kutelabhata, near Bhilai, District Durg, Chhattisgarh

IFRAME (Inline Frame)

- <iframe> is used to embed **another HTML page** inside the current page.
- src defines the page to embed.
- width and height set the size of the frame.
- frameborder="1" adds a border (use "0" for none).
- name and id help in scripting or referencing.

Example

```
<h2>IFRAME Example</h2>
<iframe frameborder="1" width="350" height="350"
src="https://www.wikipedia.org/" name="iframeme" id="iframeme">
</iframe>
```



IFRAME Example

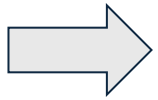


Font Tags

- <b> – Makes text **bold**.
- <i> – Makes text *italic*.
- <sup> – Superscript (e.g., x²).
- <sub> – Subscript (e.g., H₂O).

Example

```
<h2>Font Tags</h2>
<p>This is <b>bold</b> text.</p>
<p>This is <i>italic</i> text.</p>
<p>Area = a<sup>2</sup></p>
<p>Water = H<sub>2</sub>O</p>
```



Font Tags

This is **bold** text.

This is *italic* text.

Area = a^2

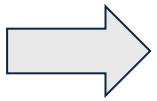
Water = H_2O

Adding Audio

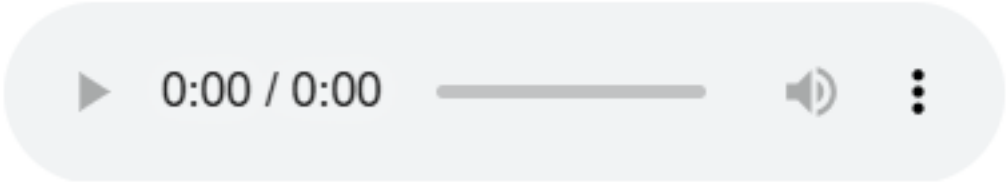
- Use <audio> tag to embed audio on the page.
- <source> specifies the file path.
- controls adds play/pause interface.
- autoplay starts playing automatically.
- loop restarts the audio after finishing.

Example

```
<h2>Audio</h2>
<audio controls>
<source src="Audio.mp3">
</audio>
```



Audio

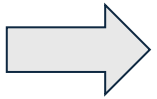


Adding Video

- Use <video> tag to embed videos.
- controls adds play/pause interface.
- autoplay starts the video automatically.
- loop keeps the video playing repeatedly.

Example

```
<h2>Video</h2>
<video width="400" height="300" controls autoplay loop>
<source src="video.mp4" type="video/mp4">
</video>
```





Thank you!

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